Bring us the hard problem

Wyspa Technologies is a US small business in Newport, RI, doing naval architecture and engineering for fast small craft since 2018: 3D laser scanning, structural analysis, composite structures, hydrofoils and appendages, and speed-power prediction. We now apply this work to unmanned surface vessels (USVs).

ENTITY	UEI	CAGE	BUSINESS TYPE
Wyspa Technologies LLC	NWJRCE69AWH9	Pending	Small business · since 2018

Core competencies

- **3D laser scanning and as-built models.** Our own FARO scanner, registered point clouds and CAD models. Scanning fed engineering on 101 projects.
- **Structural analysis and composite structures.** FEA on 31 projects (Femap/Nastran, SolidWorks Simulation), laminate schedules (ply books), and ply flattening to DXF cut files.
- **Hydrofoils and appendages.** Foils, rudders and keels on 64 projects, sized to cavitation and ventilation limits.
- **Speed-power and range at speed.** In-house prediction tools for hulls and hydrofoils, checked against ORC VPP and WinDesign on sailing craft.
- **Stability and USCG compliance.** Hydrostatics, stability and incline (GHS, Orca3D), and reports for USCG review under 46 CFR Subchapter T.
- **Ride control (simulation).** Foil-borne motion simulation with active control of ride height, pitch and roll. Not yet tested on the water.

Concept: hydrofoil USV

A 20 ft class unmanned hydrofoil craft with a proa hull (one main hull and one outrigger).

DESIGN TARGETS	
Payload	500 lb
Transit speed, foil-borne	20 kn
Range at 20 kn, 500 lb	30 nm electric; about 350 nm hybrid
Operation, foil-borne	Waves up to about 0.3 m (Sea State 2) as drawn; Sea State 3 needs deeper foils (under study)

Concept design targets, from calculation. Not yet built or tested.

Why Wyspa

- Scanning, modeling, analysis and build-ready files in one firm. No hand-off between a scan vendor and an engineering firm.
- Our own speed-power and laminate tools, so studies turn around quickly.
- A 19 ft hull available for a low-cost testbed, to check designs on the water.

Relevant experience (commercial)

- **USCG Subchapter T conversions.** A historic 12 Metre (USCG approved) and a 12-passenger Pearson 35 (structure approved, rig review pending). Scans on both; inclining test, stability and rig-loads reports on the Pearson.
- **85 ft aluminum patrol boat** (2026). Scanned on site at a working repair yard. As-built model of the outer surfaces in progress.
- **Hydrofoils for a 43 ft powercat**, with CFD of the foil system.
- **Electric-aircraft program:** 3D scans of fuselage and wings, and composite manufacturing support.
- **90+ racing-yacht projects** (TP52, Swan 42, Etchells): performance prediction, rating optimization, keels, rudders and laminates.

NAICS

541330 · 541370 · 541380 · 541690 · 541715

How to work with us

We take subcontract work packages, surge engineering and small purchase orders from USV builders, primes and government offices.

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